ADNOC GAS
HABSHAN & BAB CONTROL SYSTEMS UPGRADE (PACKAGE 1) PROJECT NO. 0405323

COMMENTS RESOLUTION SHEET

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Document Title:** | **PIPING AND INSTRUMENTATION DIAGRAM GAS SEAL SYSTEM DIAGRAM FOR 13-C-102** | | | **AGP Response Code** | | | | **Prepared By:** | VK |
| **Document No:** | **5323P1-H1-013-00-50-019** | **Revision No:** | **B** | 1. Revise and Re-submit | | 3. Approved | ✓ | **Checked By:** | DV |
| **AG Ref. No:** | **ADNOCGP-WTRAN-007148** | **Date:** | 3-Jun-2024 | 2. Approved with comments | | 4. For Information only | | **Approved By:** | YVR |

| SN | Reference | | COMPANY Comments | CONSULTANT Response | COMPANY Resolution |
|---|---|---|---|---|---|
| | Page No. | Clause No. | | | |
| 1 | - | - | No Comments. | - | |
| 2 | 1 | - | - | Following changes are made in P&ID w.r.t. Rev.B submission as per piping/instrumentation inputs:<br>1. Root valves for all PT/DPT/FT are shown in project scope cloud based on instrument input.<br>2. Control valve PDCV-3540 Tie-in location updated based on piping input. No piping tie-in required on inlet line to PDCV-3640 as these are tubing. Hence, Tie-in number TP-855A/B is updated as 13-TP-855.<br>3. Note 26 added. | |

## CONTINUATION OF NOTES :

14. CASE CONNECTION DESCRIPTION

| | |
|---|---|
| A1 - 1 - SEAL GAS SUPPLY | 2"-300# ANSI R.F. FLANGE (INTAKE & DISCH.) |
| B1 - 1 - INNER (PRIMARY) SEAL VENT | 2"-300# ANSI R.F. FLANGE (INTAKE & DISCH.) |
| C1 - 1 - OUTER (SECONDARY) SEAL VENT | 2"-300# ANSI R.F. FLANGE (INTAKE & DISCH.) |
| D1 - 1 - SEAL REFERENCE PRESSURE | 2"-300# ANSI R.F. FLANGE (INTAKE & DISCH.) |
| E1 - 1 - SEPARATION GAS INJECTION | 1-1/2"-300# ANSI RTJ FLANGE (INTAKE & DISCH.) |
| H2 - 1 - LUBE OIL VENT | 2"-150# ANSI R.F. FLANGE (INTAKE & DISCH.) |
| H1 - 1 - LUBE OIL DRAIN | 6"-150# ANSI R.F. FLANGE (IN ADPT PLT-OPP. D.E.) |
| H1 - 1 - LUBE OIL DRAIN | 3"-150# ANSI R.F. FLANGE (IN CPL. GRD ADPT-D.E.) |
| NI - 1 - BALANCE CONNECTION | 4"-300# ANSI R.F. FLANGE (INTAKE & DISCH.) |
| N2 - 1 - BALANCE GAGE CONNECTION | 1"-300# ANSI R.F. FLANGE - (DISCH. END ONLY) |
| G1 - 1 - LUBE OIL SUPPLY (NOT SHOWN) | 2"-150# R.F. ANSI FLANGE (INTAKE & DISCH.) |

15. CASE DRAINS TO BE PIPED SEPARATELY TO EDGE OF SKID.

16. BALANCE LINE CONNECTIONS TO BE BALANCE LINE DIFFERENTIAL PRESSURE GAUGE (PDI-3538) MOUNTED ON COMPRESSOR SKID.

17. SWEET FUEL GAS SEAL GAS SUPPLY SHUTOFF VALVE (SUPPLIED BY CUSTOMER) IS REQUIRED TO SHUT OFF SEAL GAS WHEN UNIT IS TRIPPED SINCE GAS SUPPLY IS NOT FROM COMPRESSOR DISCHARGE.

18. ALL INSTRUMENT TAG NUMBERS SHOWN ARE FOR THE "MP" TRAIN AND ARE PREFIXED WITH 13 - (EX. 13-PDT-3531).

19. ALL SHUTDOWNS ARE "AND" LOGIC UNLESS OTHERWISE NOTED.

20. FOR GAS SEAL DESCRIPTION OF OPERATION SEE 478-224-601.

21. EXISTING INSTRUMENT/VALVES ARE REPLACED WITH NEW INSTRUMENT/VALVES AS PER PROJECT (0405323) SCOPE.

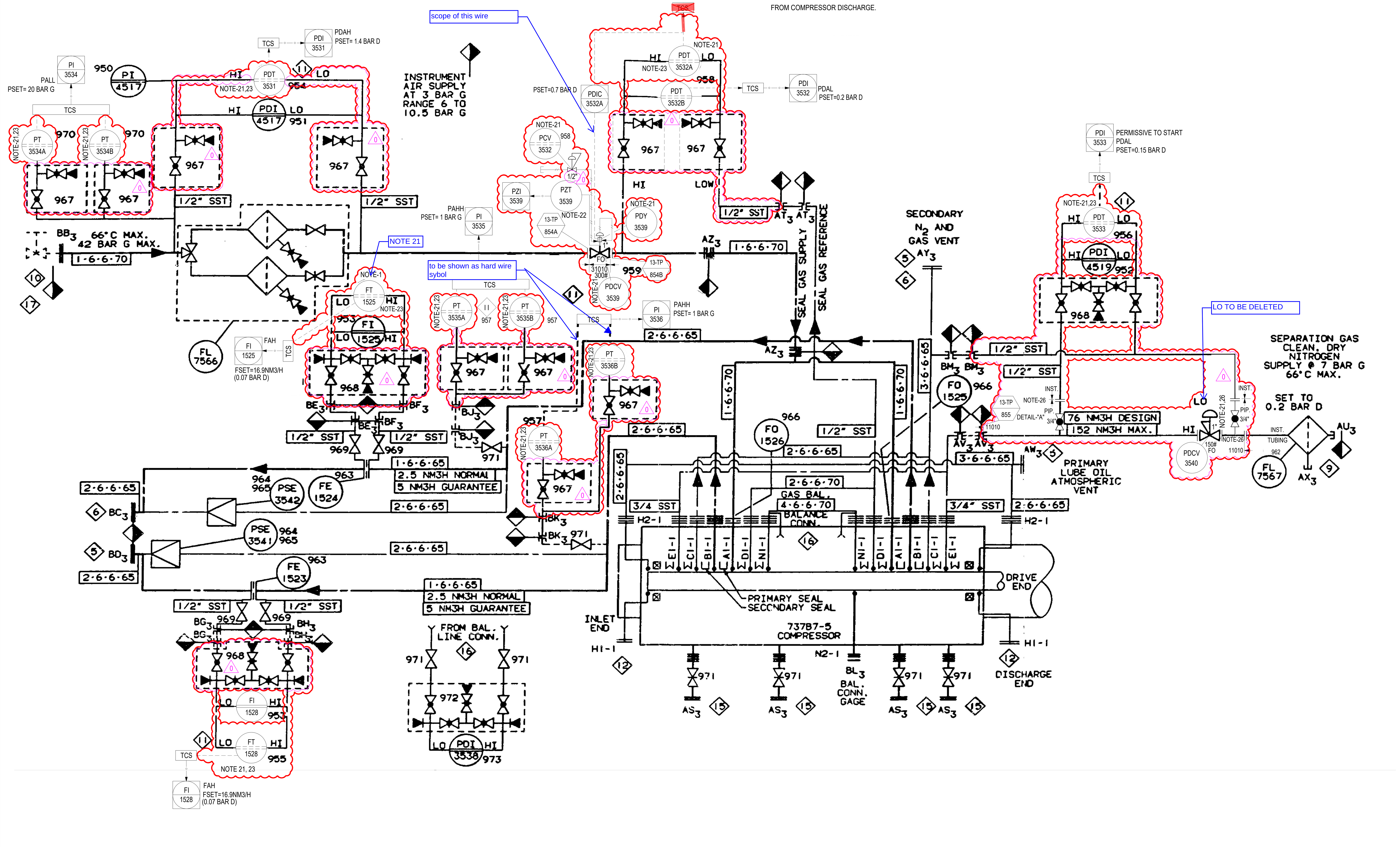
22. NEW INSTRUMENT/VALVES ARE INSTALLED AS PER PROJECT (0405323) SCOPE.

23. ALL PIPING /INSTRUMENT INTERFACE CONNECTION WILL BE AS PER HOOK-UP DIAGRAM (5323P1-H1-01-15-38-001 & 5323P1-H1-01-15-38-002).

24. THE TURBINE CONTROL SYSTEM (TCS) SHALL BE INTERFACED TO DCS THROUGH REDUNDANT MODBUS TCP/IP COMMUNICATION.

25. THE EXISTING UNIT CONTROL PANEL (UCP) RELATED TO GE RX 3I AND WOODWARD IS REPLACED WITH NEW TURBINE CONTROL SYSTEM.

26. INSTRUMENT TUBING TO BE RECONNECTED TO THE PIPING AFTER PIPE MODIFICATION. 0




## NOTES:-

1. FOR COMPONENT DESCRIPTION SEE BILL OF MATERIAL #478-224-201 AND #478-224-202.
2. FOR CUSTOMER CONNECTION LETTER DESCRIPTION SEE DRAWING 478-145-601.
3. FOR PIPING LINE CODE EXPLANATION SEE DRAWING #003-152-001.
4. FOR COMPONENT SYMBOLS AND LETTER DESIGNATIONS SEE DRAWING #478-373-601.
5. RUN VENT LINES OUTSIDE OF BUILDING AS SHOWN. DO NOT CONNECT VENTS TOGETHER EXCEPT AS SHOWN ON THIS DRAWING.
6. PRESSURE IN THIS VENT LINE MUST BE LIMITED AT ALL TIMES SO THAT MAXIMUM SEAL BACK PRESSURE DOES NOT EXCEED 0.35 BARG (5 PSIG.)
7. - LABRINTH
   - GAS SEAL
   - CARBON RING SEAL
8. A FLOW ORIFICE REQUIRES 20 PIPE DIAMETERS OF STRAIGHT PIPE UPSTREAM AND 10 PIPE DIAMETERS OF STRAIGHT PIPE DOWNSTREAM.
9. CLEAN, DRY SEPARATION NITROGEN SUPPLY@7 BAR G(100 PSIG), 2.8 BAR G (40 PSIG) MIN. FILTERED TO 3 MICRON ABSOLUTE AND 99.97% LIQUID FREE.
10. SEAL GAS SUPPLY (SWEET FUEL GAS) CLEAN, DRY FILTERED TO 10 MICRON PARTICLE (SOLIDS) AND 99.97% LIQUID FREE OF PARTICLES 3 MICRONS & LARGER.
11. FOR WIRING DETAILS SEE DRAWING #478-225-201.
12. REFERENCE LUBE OIL DIAGRAM #317-199-201.
13. SAFETY LABLE-TOXIC VAPORS (HYDROGEN SULFIDE GAS) IS TO BE AFFIXED TO GAS SEAL PANEL SEE DWG. #016-289-001.

## HOLDS:-

1. DELETED.
2. DELETED.

## LEGENDS:-

- SCOPE CLOUD
- PANEL MOUNTED
- MOUNTED BEHIND THE PANEL
- XX-TP / YYY: XX - UNIT NUMBER; TP - TIE-IN POINT; YYY - SERIAL NUMBER
- TCS: TURBINE CONTROL SYTEM INSTRUMENT (WIRED TO TCS)
- REVISION CLOUD
- LOCAL MOUNTED
- VALVE - INST. OR GLOBE - NOR. OPEN
- VALVE - INST. OR GLOBE - NOR. CLOSED
- DCS

## REFERENCE:-

1. NM-5323P1-H1-013-15-21-001 - INSTRUMENT INDEX - UNIT 13
2. 03-00-30001 TO 007 - LEGEND FOR HABSHAN-1
3. NM-5323P1-H1-01-00-40-002 - ALARM AND TRIP SETTINGS LIST - HABSHAN 1

USE THIS DRAWING FOR INFORMATION GIVEN WITHIN CLOUDED AREA

THIS DRAWING IS DEVELOPED FROM ADNOC DRAWING NO. 478-224 SHT 1/2 REV 04 (PACKAGE / VENDOR DRAWING)

| REV | DATE | DESCRIPTION | DRAWN | CHKD | APPRD | APPRD (TARGET) |
|---|---|---|---|---|---|---|
| 0 | 14.08.2024 | ISSUED FOR CONSTRUCTION (IFC) | PRA | VK/DV | YVR | WAM |
| B | 27.05.2024 | ISSUED FOR APPROVAL (IFA) | PRA | KG/DV | YVR | WAM |
| A | 13.01.2024 | ISSUED FOR COMPANY REVIEW (ICR) | PT | KG/TS | YVR | WAM |

| SCALE: | DRAWN: | CHKD: | APPRD: | APPRD (TARGET): |
|---|---|---|---|---|
| NTS | PRA | VK/DV | YVR | WAM |

أدنوك للغاز ADNOC Gas

CONTRACTOR: TARGET

ENGINEERING SUBCONTRACTOR: BILFINGER TEBODIN

PROJECT TITLE: HABSHAN & BAB CONTROL SYSTEMS UPGRADE PACKAGE 1

DRAWING TITLE: PIPING AND INSTRUMENTATION DIAGRAM GAS SEAL SYSTEM DIAGRAM FOR 13-C-102

| | | | |
|---|---|---|---|
| UNIT : 13 | | PLANT : HABSHAN-1 | |
| CONTR. PROJ. No.: AET01074.00 | | CONT. DWG. No.: 3143239 | |
| ADNOC GAS AGREEMENT No.: 4700018747 | | | |
| ADNOC GAS PROJ. No.: 0405323 | ADNOC GAS DWG. No.: 5323P1-H1-013-00-50-019 | SHEET No.: 01/01 | REV. 0 |

| | | | | |
|---|---|---|---|---|
| CONTRACT NO. 217128 | | BY | DATE | GAS SEAL SYSTEM DIAGRAM |
| TYPE INST | DRAWN | KJS | 07/06/92 | DRESSER-RAND |
| SCALE NONE | CHECKED | GMW | 07/07/92 | TURBO PRODUCTS DIVISION |
| DEPT. 63.81 | REDRAW | | | OLEAN, N.Y. USA |
| DRAFTING APPROVAL | | DJL | 07/07/92 | |
| CONTRACT APPROVAL | | JAB | 07/07/92 | |
| SYSTEM APPROVAL | | GMW | 07/07/92 | 478-224 SHT. 1 OF 2 REV. 04 |
| | | | LICENSEE | |



| | | | | |
|---|---|---|---|---|
| CONTRACT NO. 217128 | | BY | DATE | GAS SEAL SYSTEM DIAGRAM |
| TYPE INST | DRAWN | KJS | 07/06/92 | DRESSER-RAND |
| SCALE NONE | CHECKED | GMW | 07/07/92 | TURBO PRODUCTS DIVISION |
| DEPT. 63.81 | REDRAW | | | OLEAN, N.Y. USA |
| DRAFTING APPROVAL | | DJL | 07/07/92 | |
| CONTRACT APPROVAL | | JAB | 07/07/92 | |
| SYSTEM APPROVAL | | GMW | 07/07/92 | 478-224 SHT. 2 OF 2 REV. 04 |
| | | | LICENSEE | |





CAD NO.: 5323P1-H1-013-00-50-019.DWG

SHEET SIZE : A1